

CRUD Operations in PHP with PostgreSQL

CRUD means →

- C = Create** (Insert new record)
- R = Read** (Select / View records)
- U = Update** (Modify existing record)
- D = Delete** (Remove record)

Most Popular & Recommended Way

PDO + Prepared Statements + Named Parameters

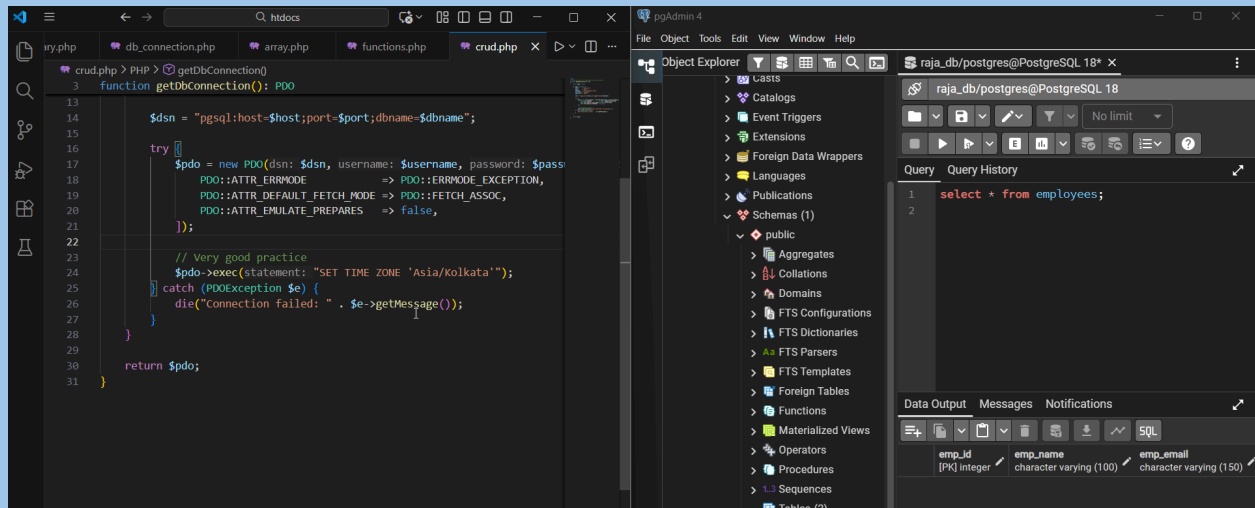
This is currently the **safest + cleanest + most modern** way

1. Example — Employee Table

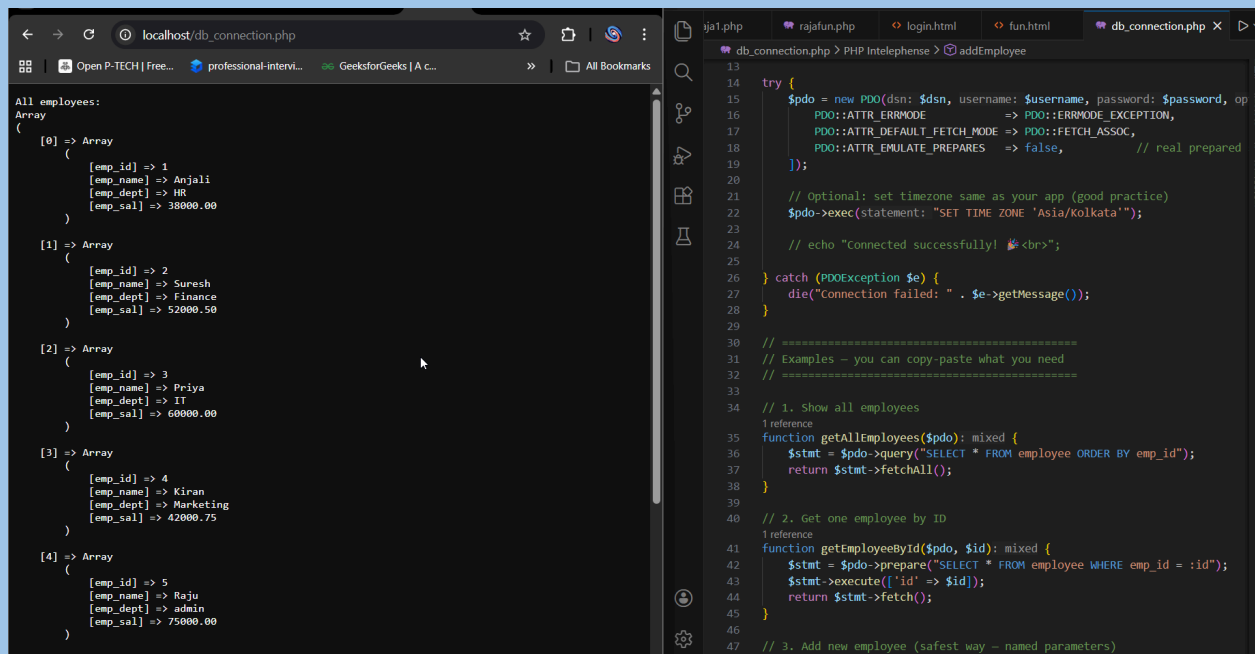
Table structure (PostgreSQL):

```
CREATE TABLE employee (  
    id          BIGSERIAL PRIMARY KEY,  
    name        VARCHAR(100) NOT NULL,  
    department  VARCHAR(50) NOT NULL,  
    salary      NUMERIC(12,2) NOT NULL,  
    created_at  TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

2. PHP Code Structure



3. CRUD Operations



```

)
[2] => Array
(
    [emp_id] => 3
    [emp_name] => Priya
    [emp_dept] => IT
    [emp_sal] => 60000.00
)
[3] => Array
(
    [emp_id] => 4
    [emp_name] => Kiran
    [emp_dept] => Marketing
    [emp_sal] => 42000.75
)
[4] => Array
(
    [emp_id] => 5
    [emp_name] => Raju
    [emp_dept] => admin
    [emp_sal] => 75000.00
)
[5] => Array
(
    [emp_id] => 6
    [emp_name] => Vamshi
    [emp_dept] => IT
    [emp_sal] => 85000.00
)
)
New employee added + emp_id = 7
Array
(
    [emp_id] => 7
    [emp_name] => Vamshi
    [emp_dept] => IT
    [emp_sal] => 85000.00
)
Salary updated
)
db_connection.php > PHP Intelephense > addEmployee
75
76
77 // 5. Delete employee
78 0 references
79 function deleteEmployee($pdo, $id): mixed {
80     $stmt = $pdo->prepare("DELETE FROM employee WHERE emp_id = :id");
81     return $stmt->execute(['id' => $id]);
82 }
83
84 // -----
85 // Quick test / demo
86 // -----
87 header(header: 'Content-Type: text/plain; charset=utf-8');
88
89 // Show all
90 echo "All employees:\n";
91 print_r(value: getAllEmployees($pdo));
92
93 // Add new
94 $newId = addEmployee($pdo, name: 'Vamshi', dept: 'IT', salary: 85000)
95 echo "\nNew employee added + emp_id = $newId\n";
96
97 // Get one
98 print_r(value: getEmployeeById($pdo, id: $newId));
99
100 // Update
101 updateSalary($pdo, id: $newId, newSalary: 92000);
102 echo "Salary updated\n";
103
104 // (uncomment if you want to test delete)
105 // deleteEmployee($pdo, $newId);
106 // echo "Deleted\n";
107
108 }
Ln 60, Col 1 Spaces: 4 UTF-8 CRLF PHP Signed out 8.4
```